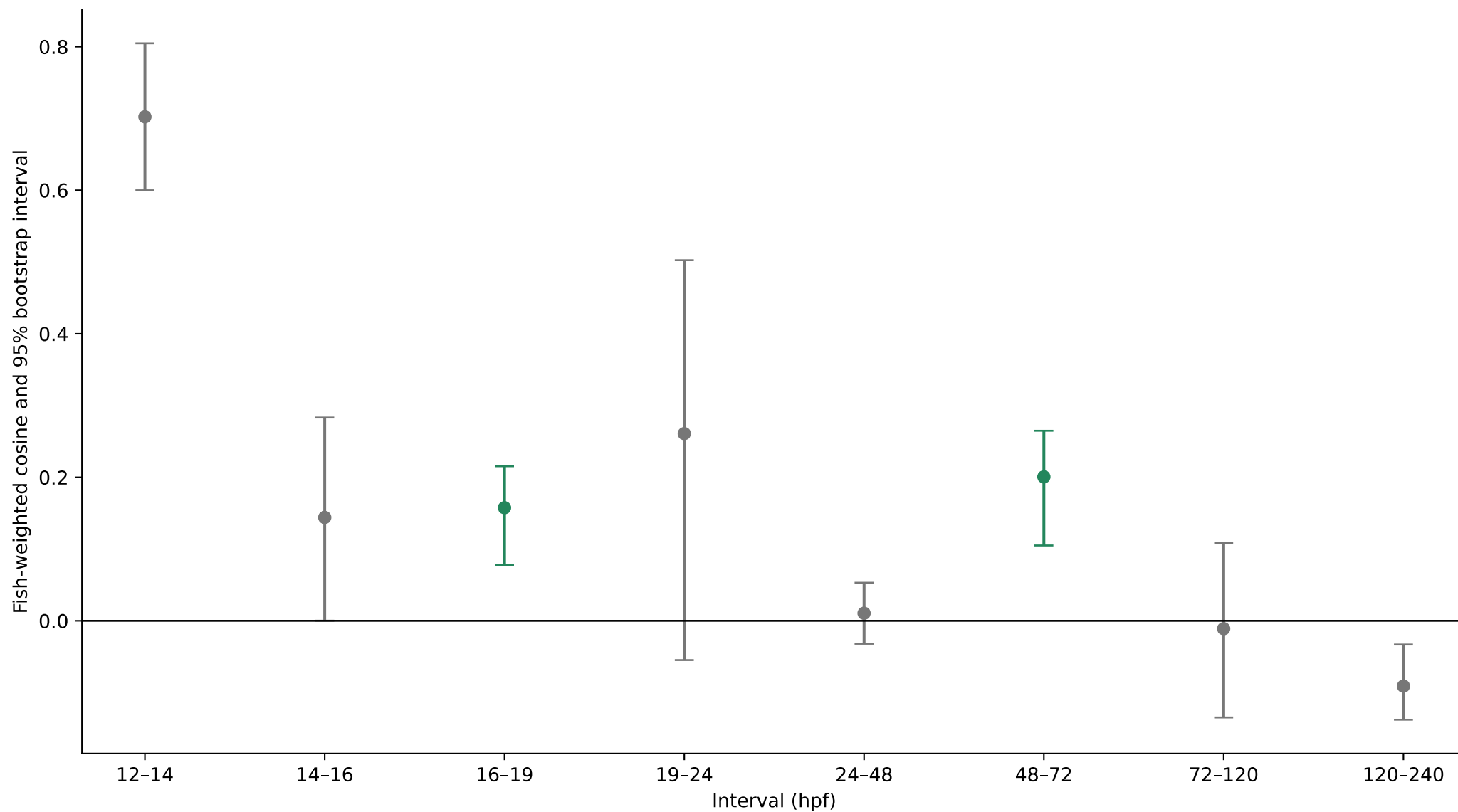


3. Directional agreement is limited to two intervals



Green: positive absolute direction, positive fish-bootstrap lower bound, and within-fish permutation FDR < 0.05 (16-19 and 48-72 hpf).
Gray: criteria not met. Late negative cosine is not positive concordance even when better than the shuffled null. Shared expression/PCA means this is complementary computational evidence.